

Trexquant — GP Alphas

7 qualifying alphas. Run 2 config: 30 seeds x 600 population x 40 generations = 738,000 expressions, tree depth capped at 5, trained 2006-2018 (includes 2008), validated 2019-2020 (COVID), tested 2021-2022. Source: gp_results_topk.csv.

How to use this

Ordered by three-window robustness first, then held-out IR — not by in-sample fitness. In run 1 the rank correlation between in-sample fitness and out-of-sample IR was **-0.70**: negative, so the in-sample ranking was actively misleading. Rows marked YES under 3-win are positive in training, validation AND the untouched 2021-2022 test block. Those are the only ones with evidence from three independent periods. Start there.

Every alpha here clears all four platform gates on LOCAL data: daily IR > 0.07, turnover < 0.5, stock count > 160, positive held-out IR. Be aware that run 1 produced 21 such alphas and none of them cleared 0.07 on Trexsim itself — the local universe is ~370 large caps against the platform's top-1000, and 212 delisted names are missing. Treat these as candidates to test, not as results.

Two things that will silently break an alpha. (1) The - 1 after every cs_rank is deliberate: Trexsim's cs_rank returns [1,2] while the GP worked in [0,1]. Removing it changes the alpha. (2) Give slow mode a generous lookback — nested operators compound their windows, so an expression with w60 inside w20 needs far more history than 60 days. If fast and slow disagree, raise the lookback first.

Copying from a PDF can insert stray line breaks into long expressions. GP_Alphas.txt next to this file has the identical strings as plain text — prefer it for pasting.

Summary

#	valid IR	test IR	3-win	daily IR	TVR	net IR	stocks	corr
1	0.97	0.32	YES	0.0840	0.088	1.15	364	0.42
2	0.82	0.39	YES	0.0952	0.053	1.42	373	0.00
3	1.07	-1.10		0.0720	0.052	1.03	350	0.43
4	0.97	-0.48		0.0874	0.032	1.34	356	0.40
5	0.66	-0.71		0.0771	0.074	1.04	364	0.44
6	0.17	-0.72		0.0750	0.042	1.13	364	0.38
7	0.01	-0.43		0.0836	0.040	1.31	356	0.50

Turnover is worth watching. The tutorial's own worked examples run 0.21-1.58; everything here sits well below that, which is the one shared trait of the run-1 alphas that failed live. If two look equally good, take the higher-TVR one.

Alpha 1 — csv rank 6, seed 3

held-out IR **0.97** | daily IR **0.0840** (gate 0.07) | TVR **0.088** (limit 0.5) | net IR 1.15 | stocks 364 (min 160) | drawdown 3.5% | max corr 0.42 | test IR 0.32 | **positive in all 3 windows**

Paste into Trexsim:

```
alpha = (ts_max(ts_zscore(ts_skew(((close - open) / at_zero2nan(open)), 5), 60), 60) + (((open - ts_delay(close, 1)) / at_zero2nan(ts_delay(close, 1))) + -((close / at_zero2nan(ts_mean(close, 20))))))
```

GP form: add(ts_max(ts_zscore(ts_skew(co, w5), w60), w60), add(gap, neg(c_ma20)))

Alpha 2 — csv rank 0, seed 24

held-out IR **0.82** | daily IR **0.0952** (gate 0.07) | TVR **0.053** (limit 0.5) | net IR 1.42 | stocks 373 (min 160) | drawdown 5.5% | max corr 0.00 | test IR 0.39 | **positive in all 3 windows**

Paste into Trexsim:

```
alpha = -((((close / at_zero2nan(ts_mean(close, 20))) * ts_sum(at_signlog((cs_rank(ts_mean(close * volume, 60)) - 1)), 20)) * (close / at_zero2nan(ts_mean(close, 20))))))
```

GP form: neg(mul(mul(c_ma20, ts_sum(signlog(size), w20)), c_ma20))

Alpha 3 — csv rank 5, seed 28

held-out IR **1.07** | daily IR **0.0720** (gate 0.07) | TVR **0.052** (limit 0.5) | net IR 1.03 | stocks 350 (min 160) | drawdown 5.4% | max corr 0.43 | test IR -1.10

Paste into Trexsim:

```
alpha = ts_median(ts_delay(ts_delay(ts_std(ts_corr_binary(((close - open) / at_zero2nan(open)), ((open - ts_delay(close, 1)) / at_zero2nan(ts_delay(close, 1)))), 10), 20), 60), 60), 60)
```

GP form: ts_median(ts_delay(ts_delay(ts_std(ts_corr(co, gap, w10), w20), w60), w60), w60)

Alpha 4 — csv rank 1, seed 9

held-out IR **0.97** | daily IR **0.0874** (gate 0.07) | TVR **0.032** (limit 0.5) | net IR 1.34 | stocks 356 (min 160) | drawdown 10.4% | max corr 0.40 | test IR -0.48

Paste into Trexsim:

```
alpha = (((close / at_zero2nan(ts_mean(close, 60))) / at_zero2nan((cs_rank(ts_mean(close * volume, 60)) - 1))) + ts_sum(ts_corr_binary(ts_skew(ts_corr_binary(ret1, ret1_spx, 60), 60), ts_mean(((open - ts_delay(close, 1)) / at_zero2nan(ts_delay(close, 1))), 10), 60), 60))
```

GP form: add(div(c_ma60, size), ts_sum(ts_corr(ts_skew(beta60, w60), ts_mean(gap, w10), w60), w60))

Alpha 5 — csv rank 7, seed 3

held-out IR **0.66** | daily IR **0.0771** (gate 0.07) | TVR **0.074** (limit 0.5) | net IR 1.04 | stocks 364 (min 160) | drawdown 3.1% | max corr 0.44 | test IR -0.71

Paste into Trexsim:

```
alpha = (ts_max(ts_zscore(ts_std(ts_skew(((close - open) / at_zero2nan(open)), 5), 5), 60), 60) + (((open - ts_delay(close, 1)) / at_zero2nan(ts_delay(close, 1))) + -((close / at_zero2nan(ts_mean(close, 20))))))
```

GP form: add(ts_max(ts_zscore(ts_std(ts_skew(co, w5), w5), w60), w60), add(gap, neg(c_ma20)))

Alpha 6 — csv rank 4, seed 23

held-out IR **0.17** | daily IR **0.0750** (gate 0.07) | TVR **0.042** (limit 0.5) | net IR 1.13 | stocks 364 (min 160) | drawdown 9.5% | max corr 0.38 | test IR -0.72

Paste into Trexsim:

```
alpha = ts_mean(ts_max(ts_mean_exp(((ts_corr_binary(ret1, ret1_spx, 60) - ts_delay(ts_corr_binary(ret1, ret1_spx, 60), 5)) - ((close * volume) / at_zero2nan(ts_mean(close * volume, 60)))), 10, 0.5), 60), 10)
```

GP form: `ts_mean(ts_max(ts_decay(sub(ts_delta(beta60, w5), dv_adv), w_id(w10)), w60), w_id(w10))`

Alpha 7 — csv rank 2, seed 9

held-out IR **0.01** | daily IR **0.0836** (gate 0.07) | TVR **0.040** (limit 0.5) | net IR 1.31 | stocks 356 (min 160) | drawdown 18.1% | max corr 0.50 | test IR -0.43

Paste into Trexsim:

```
alpha = (((close / at_zero2nan(ts_mean(close, 60))) / at_zero2nan((cs_rank(ts_mean(close * volume, 60)) - 1))) + ts_mean(ts_min(ts_skew(ts_max(ts_corr_binary(ret1, ret1_spx, 60), 60), 60), 60), 5))
```

GP form: `add(div(c_ma60, size), ts_mean(ts_min(ts_skew(ts_max(beta60, w60), w60), w60), w_id(w_id(w5))))`